

Markov Chain Simulation and Convergence Analysis

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Introduction

This report investigates the long-run behavior of a Markov chain using both analytical methods and simulation.

The objective is to study convergence to the steady-state distribution, examine the effect of initial conditions, and validate theoretical results through computational experiments.

Methodology

Three approaches were used:

1. Matrix method: computing aP^k
2. Steady-state equation: solving $\pi P = \pi$
3. Simulation: sampling random transitions over multiple runs

The results from these approaches are compared to study convergence behavior.

Results

The following figures illustrate convergence behavior, steady-state stability, and the effect of initial conditions.

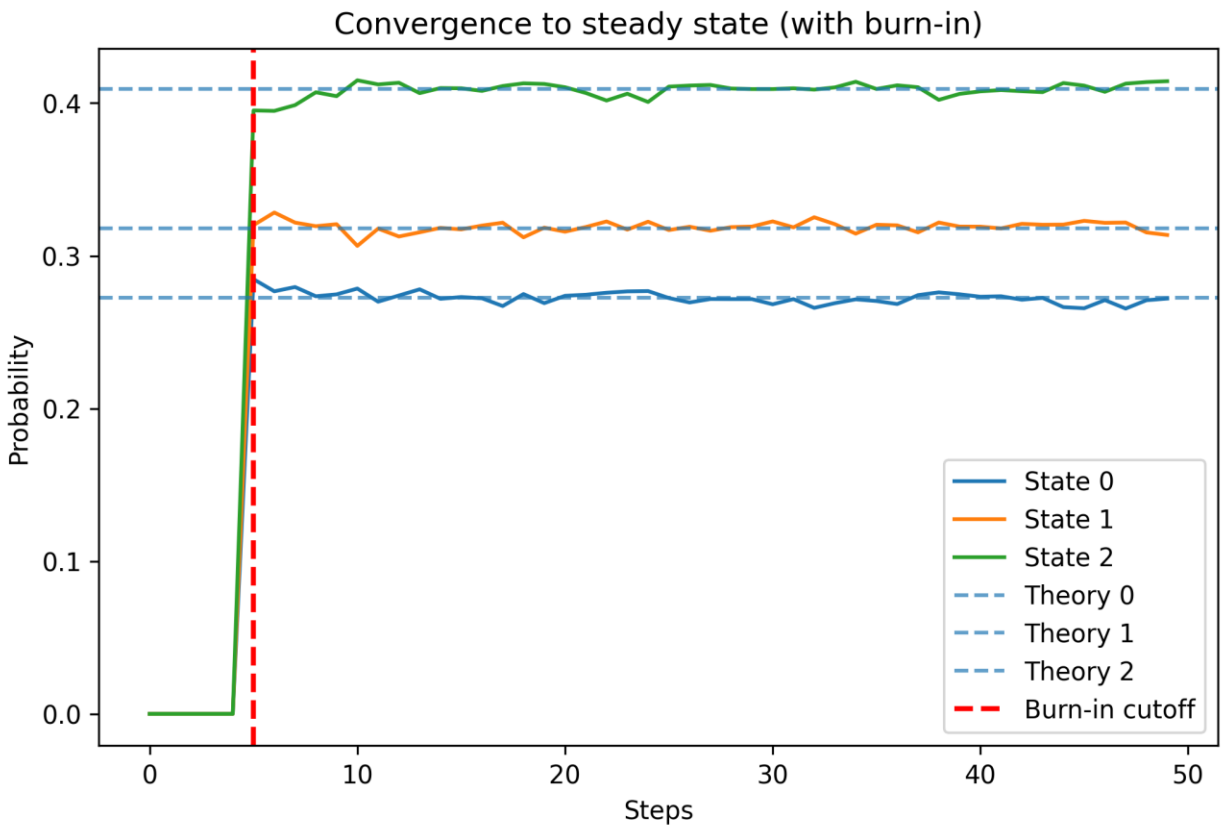


Figure 1: Convergence of state probabilities to steady-state values. The vertical dashed line indicates the burn-in period.



Figure 2: Zoomed view of steady-state behavior. Small fluctuations around equilibrium are due to randomness.

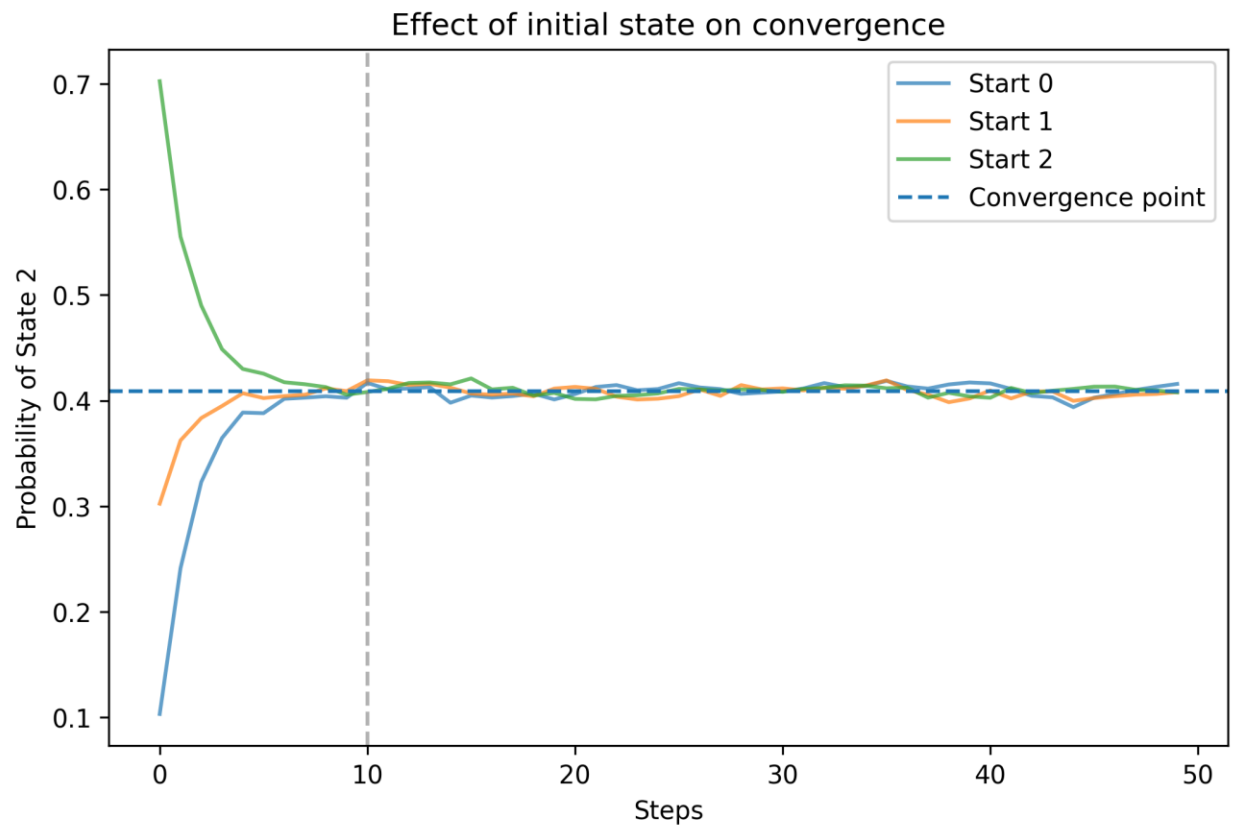


Figure 3: Effect of different initial states. All trajectories converge to the same steady-state distribution.

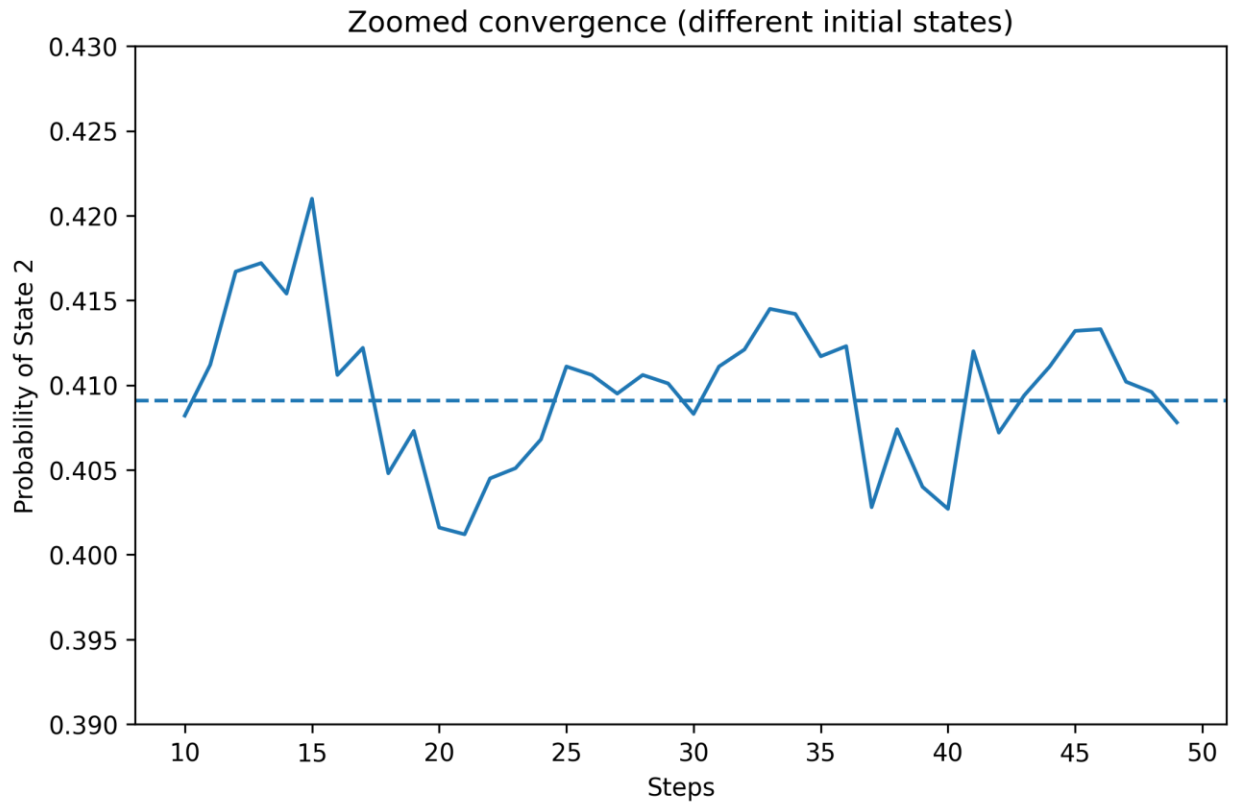


Figure 4: Zoomed view of steady-state behavior for a representative trajectory.

Discussion

The results show that the Markov chain in this example converges to a steady-state distribution. In general, convergence depends on properties of the transition matrix, and the speed of convergence may vary across different systems.

Although the simulation exhibits fluctuations due to randomness, the long-run values match theoretical predictions closely.

The convergence occurs regardless of the initial state, confirming that the system is independent of initial conditions.

The burn-in period captures transient behavior, after which the system stabilizes.